Predictive Refinement Across Transformer Depth: Signal, Fluctuation, Information Exposure, and Predictive Order

Paper 0.5 — Residual / Layer-Composition Research Line

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Abstract

Once relevant context is accessible, what does Transformer depth compute? In a synthetic micro-mechanistic regime, we independently control predictive order, nuisance, information exposure, raw length, and dependency span. Correct-target signal grows despite finite or increasing fluctuation, and negative cross-layer covariance supports later repair; monotonic denoising, entropy reduction, and globally composed linear response are falsified. Reachability is necessary but insufficient: arbitrary restriction breaks learned routing, whereas selectively masking nuisance keys preserves perfect accuracy and reduces margin variance. A matched three-seed stress test makes predictive order the binding competence variable: two parameter-matched architectures solve order two across every tested raw length ($n \leq 24$), nuisance count ($k \leq 32$), dependency span ($s \leq 32$), and generator family, yet both fail from order three onward. A fourfold training budget is required to make all baseline seeds competent; this is acquisition rescue, not evidence that training enlarges representational capacity. Approximate-budget recurrent controls falsify a broad recurrent disadvantage: GRU is perfect and LSTM strong, although vanilla RNN shows a transport limitation. The supported account is *nonlinear predictive refinement*: accessible evidence is iteratively transformed until correct-target contrast dominates realization-dependent fluctuation and competing alternatives. Pretrained diagnostics and legacy pilots remain secondary.

1 Introduction

Once relevant context is accessible, what computation does Transformer depth perform? Self-attention can make distant information available at a prediction position, but access does not imply that the model has extracted the right predictive consequence. Conversely, a long input need not demand many refinement steps when a small sufficient subset already determines the target. This distinction is obscured when context length, transport distance, nuisance, and integration complexity vary together.

We study this question in a synthetic micro-mechanistic regime where predictive structure, nuisance, and information accessibility can be independently controlled; our claims concern these controlled dynamics rather than universal behavior of large pretrained language models. We keep raw length n, indispensable-token span s, nuisance count k, and predictive order p^* separate from the outset.

We call *predictive refinement* the depthwise process by which correct-target evidence becomes increasingly dominant over realization-dependent fluctuation and competing alternatives. The definition is behavioral and trajectory-based. It does not assume that hidden-state variance shrinks, that entropy falls, that every layer improves the answer, or that a visually recurrent attention motif is causal.

We use controlled generators because the target, sufficient subset, nuisance relation, and attention-mask graph are known exactly. Our contributions are:

1. a correctness-aware signal/fluctuation framework with an exact cross-layer repair decomposition;
2. causal tests separating information reachability, learned routing, arbitrary restriction, and selective nuisance exclusion;
3. matched controls separating raw length n, dependency span s, nuisance count k, and predictive order p^*;
4. a three-seed, parameter-matched stress frontier over predictive order, raw length, nuisance, span, and training budget, plus a matched Transformer/RNN/GRU/LSTM sensitivity comparison;
5. a claim audit that retains falsifications, failed competence cells, and weak external validity.

2 Related work

LayerDrop and conditional-depth methods show that Transformer computation can be pruned or allocated non-uniformly [2, 3, 4]; early-exit work and the tuned lens decode intermediate predictions [5]. We instead ask which controlled predictive variables change the formation and stabilization of the correct answer, without training an exit policy.

Mechanistic work studies induction heads, residual streams, causal tracing, activation replacement, and FF layers as associative memories [6, 7, 8]. These motivate our interventions but not our conclusions: component magnitude, motif recurrence, and causal utility are measured separately, and localization is not treated as semantic storage.

Long-context studies document position effects, irrelevant-context degradation, and distractibility [9, 10]. Sparse/local attention establishes graph constraints, while retrieval noise can alter behavior. Our mask experiments distinguish formal reachability from learned functional transport and compare arbitrary truncation with selective key exclusion.

Residual networks motivate dynamical and Jacobian views [12, 13]; representation and information analyses motivate separation-to-fluctuation statistics. We test local and cumulative linearization directly rather than assuming a smooth global flow. Activation-patching methodology further warns that metrics and corruption choices can change localization results [11]. Finally, recurrence has a sequential path whose length grows with source–target distance, whereas full attention provides a direct edge within each layer [1, 14, 15]. Our matched synthetic comparison concerns sensitivity along controlled axes, not universal architecture superiority; modern selective state-space models remain outside scope [16].

Prior work separately studies intermediate predictions, depth allocation, context interference, residual dynamics, and recurrence. We jointly control predictive order, raw length, dependency span, nuisance, reachability, correctness, and architecture, allowing several common refinement interpretations to be directly falsified. The separating results are length $\neq$ predictive order, reachability $\neq$ functional routing, stability $\neq$ variance contraction, and local linearity $\neq$ global linear composition. We do not claim novelty for the generic observation that predictions change with depth.

3 Controlled framework and theory

3.1 Signal, fluctuation, and repair

A residual block is

$$r_{l+1} = r_l + F_l(r_l). \tag{1}$$

For realization i, define correct-target margin

$$m_{li} = z_{li}(Y) - \max_{y \neq Y} z_{li}(y) = S_l + \eta_{li}, \tag{2}$$

where $S_l = \mathbb{E}_i[m_{li}]$, $N_l^2 = \mathrm{Var}_i(m_{li})$, and

$$\Gamma_l = \frac{S_l}{\sqrt{N_l^2 + \epsilon}}. \tag{3}$$

Positive margin is exactly correct top-1 behavior. We report S_l and N_l^2 beside Γ_l because a large ratio can arise from a nearly zero denominator.

Write a boundary update as $\Delta m_{li} = \mu_l + \varepsilon_{li}$. Then

$$m_{Li} = m_{0i} + \sum_l \mu_l + \sum_l \varepsilon_{li}, \tag{4}$$

$$\mathrm{Var}(\sum_l \varepsilon_l) = \sum_l \sigma_l^2 + 2 \sum_{l<j} \mathrm{Cov}(\varepsilon_l, \varepsilon_j). \tag{5}$$

Negative off-diagonal covariance means later updates cancel part of earlier realization-specific deviation. This *repair* need not be monotone layer by layer. Our reported "summed off-diagonal covariance" denotes $2\sum_{l<j} \mathrm{Cov}(\varepsilon_l, \varepsilon_j)$ and therefore already includes the factor of two.

3.2 Predictive order and controlled axes

For token-index set A, define

$$p^* = \min_A |A| \quad \text{s.t.} \quad H(Y \mid X_A) \leq \delta, \tag{6}$$

with fixed audit threshold δ. Intuitively, p^* is the smallest number of jointly available input variables sufficient to determine the target to the chosen uncertainty tolerance. Raw length n counts surface tokens; dependency span s

measures distance among indispensable tokens; nuisance k counts tokens satisfying $Y \perp d \mid g$. These quantities are crossed rather than used interchangeably. Necessary modular generators have zero target MI for prohibited proper subsets and two bits for the full pattern.

3.3 Local linearization and information access

For a paired displacement,

$$\delta r_{l+1} = J_l \delta r_l + O(\|\delta r_l\|^2). \tag{7}$$

We test one-step and cumulative JVP predictions; the full propagation derivation is in the appendix. Failure of the frozen cumulative product cautions against treating Transformer depth as a single globally valid tangent flow: local linear response is informative, but the relevant tangent geometry changes substantially along the trajectory. With local attention window w, $Lw \geq s$ is a graph reachability condition in our mask, not a learning guarantee. Attention makes a distinction accessible; composed SA/FF transformations determine whether and how it affects the decision.

4 Experimental setup

The primary generators use four balanced targets, role-specific value tokens, same-vocabulary nuisance, disjoint deterministic train/test families, and exact information audits. Dataset V2 contains pair, three-token, sparse, functor, and nested rules. Every residual/SA/FF boundary records target probability, margin, rank, top-1, entropy, first/stable decision depth, reversals, and covariance statistics. Causal interventions include local windows, key-selective masks, component/head ablations, and matched activation donors.

The original phase grid evaluates $p^* \in \{1, 2, 3, 4, 6, 8\}$ over depths 2, 4, 8, 12, 16, and 24, widths 32 and 64 where compute-feasible, a 400-update base budget, and 2x/4x budgets at selected difficult cells. Its surface uses seed 11 and is retained as motivation. The confirmatory stress grid uses three seeds and a parameter-matched pair: L2/W64/H4 (85,504 parameters) and L8/W32/H4 (77,632; 9.2% fewer). It varies one data axis at a time around $p^* = 2$, tests $p^* \in \{1, 2, 3, 4, 6, 8, 12\}$, $n \leq 24$, $k \leq 32$, and $s \leq 32$, and compares exact-prefix 1x/2x/4x training budgets. The primary/secondary competence thresholds are fixed analysis gates of 0.80/0.90, not claimed as preregistered. The recurrent comparison holds generator, optimizer family, examples, update count, vocabulary, target entropy, and training tokens fixed across Transformer, vanilla RNN, GRU, and LSTM; parameter matching is approximate. Internal RNN time is not equated with Transformer depth.

5 Results

5.1 Result 1: depth builds correct predictive contrast

Across the primary three-generator matrix, evaluated on two small Transformer configurations and seeds 11/23 after each family clears the 0.80 gate, eight-nuisance accuracy is 1.000. The independent variable is residual boundary under crossed nuisance realization. Mean correct margin changes from -32.02 initially to $+10.51$ finally, while target probability rises from 1.4×10^{-10} to 0.9998. Absolute margin variance increases from zero to 1.89, so reliability comes from signal growth outpacing fluctuation rather than denominator collapse. Entropy is largely independent of margin (correlation 0.081) and may rise while correctness forms.

Only 7/27 family–seed trajectories contract monotonically; 11 expand then contract and the rest oscillate, contract late, or fail to contract. Mean first/stable boundaries increase modestly with nuisance, but settling delay remains short. Transformer depth is therefore not a sequence of independent estimators or guaranteed denoisers.

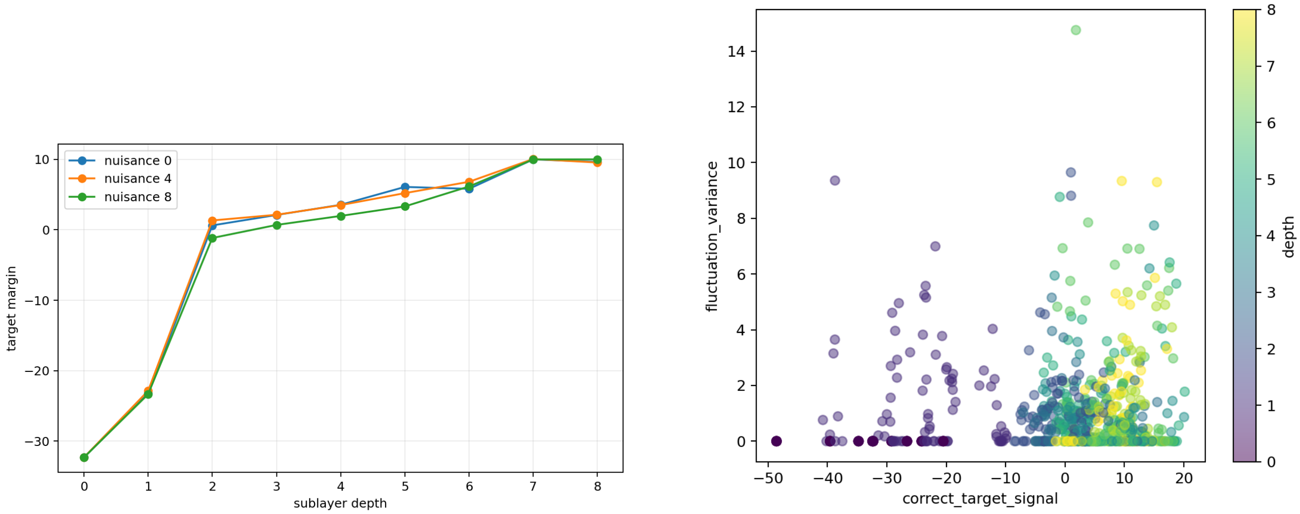

Figure 1: Competent controlled Transformer families across residual depth (two architectures, two seeds): correct-target margin and the signal–fluctuation trajectory. Correct contrast grows despite finite and sometimes increasing nuisance-conditioned variation.

5.2 Result 2: refinement includes correlated repair and is globally nonlinear

Mean boundary progress is positive, yet only 73.7% of family/seed/noise transition cells improve on average. Median summed marginal update variance is 3.55, while median summed off-diagonal covariance is −2.12, leaving cumulative fluctuation variance 0.51. Later transformations therefore repair a substantial portion of earlier deviation.

Local JVPs predict nuisance and signal directions well over one block. Across 85 Dataset V2 pairs filtered by held-out behavioral correctness, cumulative horizon is the independent variable: directional cosine drops from 0.782 at one block to 0.552 at two and 0.335 at four; relative error rises from 0.624 to 1.047. Piecewise re-linearization improves on the frozen product. Local directional structure is real, but a single frozen globally linear composition is falsified.

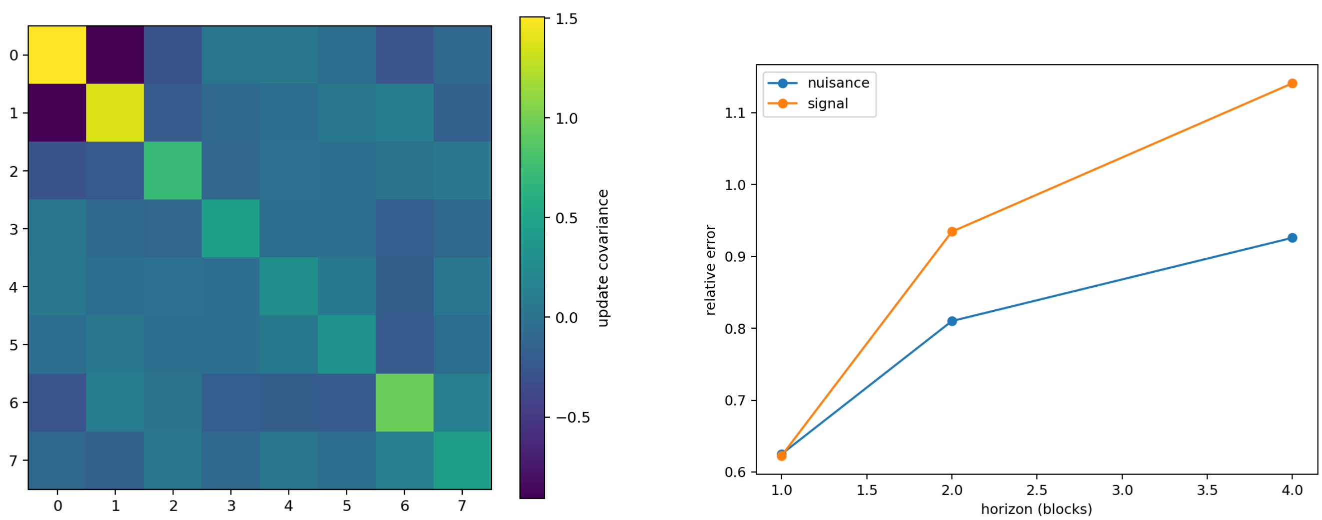

Figure 2: Competence-filtered controlled Transformers: cross-boundary update covariance and cumulative-JVP error versus horizon. Negative covariance implements repair; frozen linearization deteriorates with horizon.

5.3 Result 3: information exposure changes refinement burden

The two-seed controlled Transformer window grid varies dependency span, attention window, and exposure during training while retaining the 0.80 competence gate for mechanistic interpretation. The local-window mask verifies exact graph reachability. Unreachable cells remain near chance, but reachability is not sufficient: one reachable $(s, w, L) = (8, 2, 4)$ cell attains only 0.078 accuracy. Full-trained models evaluated under arbitrary truncation average only 0.411 in the nominal signal-reachable/nuisance-unreachable regime, whereas locally trained restricted models reach 1.000. Learned routing and exposure during training matter.

Selective masking provides the cleaner causal contrast. Blocking nuisance keys preserves 1.000 accuracy, increases mean target margin by 0.229 logits, and reduces nuisance-conditioned margin variance from 1.864 to 1.109 (40.5%). Blocking predictive keys collapses accuracy to 0.249. SA and FF both change variance, and median repair fraction is 0.812. Less context helps only when exclusion respects the learned signal path.

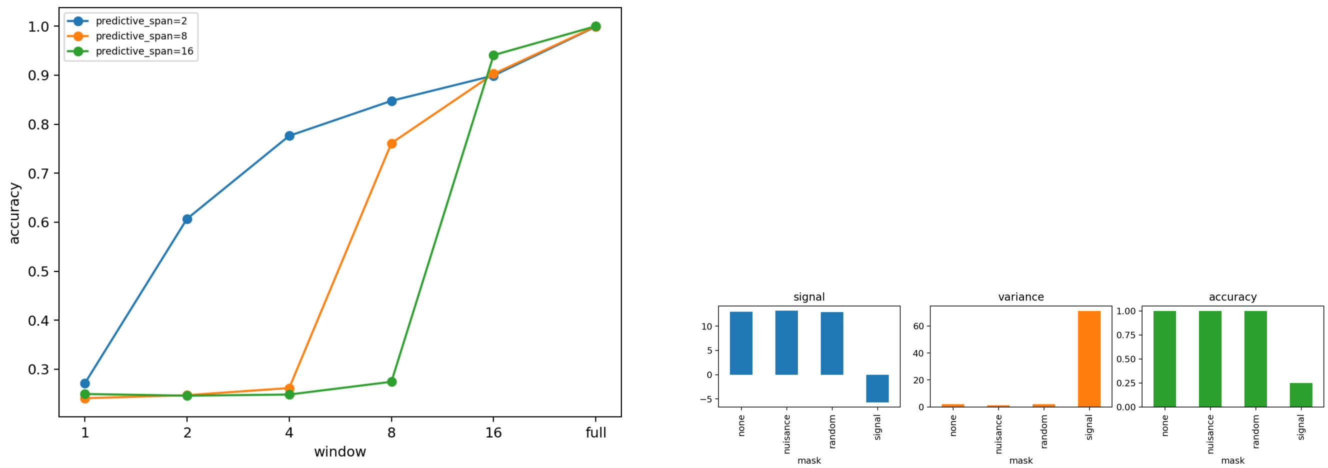

Figure 3: Two-seed controlled Transformer window and key-mask interventions. Reachability does not guarantee learned transport; on competent full-context models, selective nuisance exclusion reduces burden without deleting signal.

5.4 Result 4: predictive order, not surface burden, binds competence

In the competence-gated nested study, redundant accuracy stays 0.961–1.000 and supportive accuracy 0.994–1.000 through length eight. Their stable decision boundaries do not increase consistently; the descriptive raw-length fit has $R^2 = 0.016$. Necessary-pattern accuracy in the initial smoke falls from 0.996 at order two to 0.500, 0.322, and 0.240 at orders four, six, and eight. Those failures motivated rather than answered the final phase experiment.

Across 14 models, orders 1–2 reach the 0.80 gate at base budget. At depth 8/width 64, order 3 rises from 0.634 at 1x to 0.927/0.972 at 2x/4x, while order 4 reaches only 0.491. Best accuracies for orders 4/6/8 are 0.507/0.387/0.391. Depth is non-monotonic: order 2 is perfect at depth 8/width 64, falls below 0.60 at depths 12/16, and recovers at depth 24. Interaction fits differ only modestly. Thus predictive order changes competence, but no order-to-minimum-depth law is identified; width and optimization are inseparable in this grid.Architectural depth is not an independently monotonic resource under fixed training conditions: depth 8 is competent at order two, depths 12/16 are poor, and depth 24 recovers. Optimization also moves the frontier rather than merely sharpening an already fixed boundary: at depth 8/width 64, order-three accuracy rises from 0.634 to 0.927 and 0.972 under 1x, 2x, and 4x training. These are one-seed phase estimates and do not identify a universal depth law.

The matched stress test resolves the controlled comparison. At the nominal 1x budget only two of six architecture/seed baselines are competent; exact-prefix 2x training rescues five, and 4x rescues all six to 1.000 held-out accuracy. We therefore interpret factor slopes only at 4x. Both architectures have the same measured frontier, $p^*_{\max} = 2$: their minimum-seed accuracy falls below the 0.80 gate at $p^* = 3$ and remains below it through 12. In contrast, every seed is perfect for every tested raw length through $n = 24$ and nuisance count through $k = 32$. Span remains competent through $s = 32$ (minimum 0.969), and contiguous, skip-gram, and necessary-pattern generators all pass. Mean stable boundary changes only from 1.39 to 1.42 layers over $k = 0 \to 32$ for L2/W64 and from 2.76 to 2.96 for L8/W32. Within this bounded regime, surface extent and nuisance neither reduce accuracy nor materially delay stabilization; jointly necessary predictive variables do.

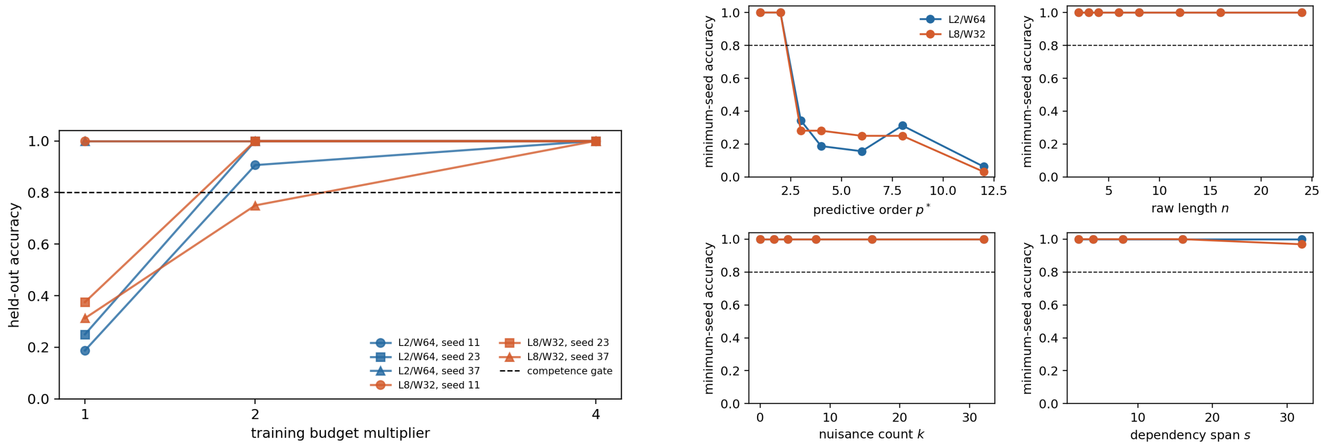

Figure 4: Three-seed matched stress test. Left: exact-prefix budget rescue of the $p^* = 2$ baseline; the full interpretation gate opens only at 4x. Right: minimum-seed accuracy at 4x. Predictive order binds at two, whereas all tested length, nuisance, and span cells remain competent.

Training rescue establishes that both architectures can acquire the baseline under this optimizer and exposure; it does not show that additional updates create a larger representational class. Nor does the shared frontier prove depth and width interchangeable beyond these two approximately matched models. It shows only that eight narrow layers do not extend the measured order frontier relative to two wider layers here.

At fixed $p^* = 2$, the Transformer and GRU remain at 1.000 across lengths 2–32 and spans 1–64. Vanilla RNN falls to 0.547–0.656 at lengths 16–32 and 0.578 at spans 32–64; LSTM falls only to 0.898 at length 32 and remains perfect across span. Conversely, GRU reaches 1.000 for every predictive order through eight, LSTM reaches 0.750 at order eight, and the Transformer reaches 0.273. Therefore only the vanilla-RNN transport contrast is supported; a conventional-recurrence disadvantage and universal Transformer superiority are falsified.The matched models contain approximately 40.9k Transformer, 16.6k vanilla-RNN, 33.3k GRU, and 41.6k LSTM parameters; all receive 1,200 updates and 12.288M training tokens. Parameter matching is approximate, and the shared optimizer/batch schedule is a controlled comparison rather than architecture-specific exhaustive tuning. The GRU result shows that predictive refinement is not Transformer-specific. Gated recurrence can realize the controlled integration task extremely effectively, indicating that the broader computational principle may be iterative state transformation rather than self-attention itself. The vanilla RNN shows a transport limitation; the GRU does not, the LSTM is strong, and the Transformer is not universally superior on these generators.

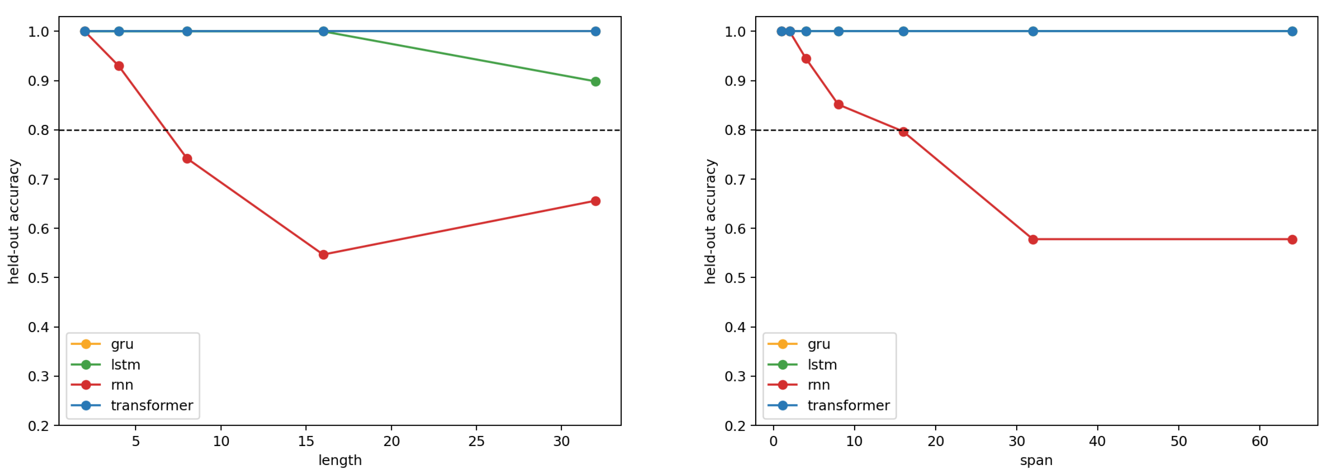

Figure 5: Two-seed Transformer/RNN/GRU/LSTM sensitivity to raw length and indispensable-token span at matched 1,200-update/12.288M-token exposure. Parameter matching is approximate; predictive-order/nuisance curves are reported in the appendix.

Target-preserving nested replacement supplies complementary causal evidence. Short-pattern donors are less damaging than nonequivalent donors, while block/head overlap remains partial. This supports reuse of a short computation under competent extension, not a fixed semantic circuit or a general hierarchy.

5.5 Hypothesis audit

Hypothesis	Status	Decisive evidence
Correct signal grows relative to fluctuation	supported	margin/SNR trajectory
Monotonic nuisance suppression; entropy reduction	falsified	non-monotone paths; entropy independence
Later correlated repair	supported	negative off-diagonal covariance
Global frozen-Jacobian composition	falsified	horizon-dependent JVP failure
Reachability is sufficient	falsified	reachable optimization failure
Selective nuisance exclusion helps	supported	paired key masking
Raw length controls competence/settling depth	not supported	three-seed $n \leq 24$ null
Predictive order binds the measured competence frontier	supported	three-seed $p^*_{\max} = 2$
Nuisance necessarily delays decisions	not supported	competent $k \leq 32$ null
Stable head motifs explain utility	not supported	repeated motif–causal null
RNNs are more length/span sensitive	falsified broadly	matched controlled curves

Table 1: Claims are classified as supported, falsified, not supported, or unresolved; failed competence cells never become positive mechanism evidence.

6 Discussion and limitations

Information access versus refinement. Attention shortens the graph path by which evidence can reach the query, but useful prediction still requires learned nonlinear transformations. Width, depth, and training may trade off, yet an observed failure identifies none of them until competence recovers somewhere on the controlled surface.

Selective exposure. More context is neither inherently better nor worse. Arbitrary restriction can break trained routing; selective exclusion of known nuisance can reduce fluctuation without deleting signal. This distinction matters for sparse-context systems, but the synthetic oracle labels do not solve nuisance identification in natural text.

Architecture. The Transformer is invariant to tested length/span while vanilla RNN degrades, but GRU is perfect and LSTM nearly perfect; a general recurrent disadvantage is falsified. The comparison is limited to conventional unidirectional RNN/GRU/LSTM baselines and a small Transformer. It excludes state-space models, recurrent attention, and extensive architecture-specific tuning.

Three implications. Predictive refinement is architecture-independent in principle: the GRU realizes it through gated recurrence. Information exposure is selective, not monotonic: more context is neither always better nor always worse. Predictive order is distinct from surface burden: once the order-two computation is acquired, length, nuisance, and span do not move the tested frontier. Depth is likewise not a monotonic resource within a trajectory or across trained architectures. More layers do not guarantee a higher order frontier under matched parameter count and optimization.

The main limitations are synthetic modular generators, small custom models, high-order failure that may still combine optimization and capacity, only two architectures in the three-seed stress comparison, a one-seed legacy phase surface, approximate recurrent parameter matching, limited recurrent families, no state-space comparison, limited semantic complexity, and weak pretrained external validity. The tested maxima $n = 24$, $k = 32$, and $s = 32$ are lower bounds on frontiers, not saturation points. No universal scaling law, general Transformer superiority, or semantic-abstraction claim is made. Head roles are narrow task-conditioned causal summaries. Pretrained Pythia/Qwen results remain appendix diagnostics because the true training families and sufficient subsets are unknown.

7 Conclusion

Transformer depth is not well described as monotonic denoising or computation proportional to context length. In these controlled tasks, depth performs nonlinear predictive refinement: correct evidence grows relative to realization-dependent fluctuation and competitors, later layers repair earlier deviations, and information exposure constrains but does not determine the learned computation. The matched stress frontier sharpens the separation: order two is acquired across every tested surface extension, while order three is not reliably acquired by either parameter-matched architecture. Training budget controls whether this comparison is observable, but does not by itself identify the representational limit. The resulting theory is deliberately conditional and competence-gated: structured predictive distinctions, not token count alone, determine the burden that remains after context becomes accessible.

A Generator definitions and information audits

All controlled targets have four balanced classes. For the no-shortcut family, active values X_j are independent and uniform on $\mathbb{Z}_4$ and

$$Y = \left(\sum_{j=1}^{p^*} X_j \right) \bmod 4.$$

For $p^* > 1$, any prohibited proper subset is independent of Y: conditioning on that subset leaves at least one independent uniform summand, which one-time-pads the modular total. Consequently $I(X_A; Y) = 0$ for every proper $A \subsetneq \{1, \ldots, p^*\}$, whereas $I(X_1, \ldots, X_{p^*}; Y) = H(Y) = 2$ bits. Orders up to four are audited exhaustively; orders six and eight use this analytic identity plus exhaustive singleton/full-pattern checks. Role-specific token pools prevent a bag-of-token-count shortcut. Train and test examples use disjoint deterministic identity ranges.

The length controls alter distinct variables. Redundant extensions retain a sufficient two-token core and add conditionally irrelevant tokens. Supportive extensions add target-compatible evidence and therefore intentionally have informative singleton cues. Necessary extensions increase p^*. The final phase generator matches the latent values and target while varying raw length n, span s, and nuisance count k.

B Training and configuration

The primary controlled Transformer is a pre-norm causal residual model with tied input/output embeddings, multi-head self-attention, GELU feed-forward blocks, and a shared diagnostic decoder. Optimization uses AdamW, gradient-norm clipping at one, balanced online generation, fixed seeds, and held-out deterministic families. The predictive-order phase grid crosses the tracked depths, widths, and training budgets in `configs/paper05/predictive_order_phase.json`. The compute-feasible width subset and all omitted grid cells are explicit in the manifest; a blank minimum-depth/width cell means that competence was not reached, not that the boundary lies at infinity.

The recurrent comparison uses vanilla RNN, GRU, and LSTM baselines with the same vocabulary, generated examples, optimizer family, update count, batch size, target entropy, and token budget as its Transformer baseline. Exact parameter counts are reported rather than described as perfectly equal. Architecture-specific tuning was not enlarged after seeing test results. Recurrent time t and Transformer layer l are never treated as commensurate compute units.

C Statistical details and failed scaling fits

The scientific unit is a held-out predictive family, model seed, and architecture cell rather than correlated token occurrences. Competence thresholds are fixed at 0.80 (primary) and 0.90 (secondary). Internal decision-depth, reuse, and head analyses exclude architecture/order strata below the primary threshold. Descriptive regressions include p^*, n, s, k, depth, width, training steps, and preregistered interactions. Constant, linear, logarithmic, power, threshold, and saturating descriptions are compared where variation permits; no power law is promoted unless it clearly improves held-out error and information criteria. Existing fits do not meet that standard.

Dataset V2 uses family/seed bootstrap intervals. Group 2.5 uses architecture-cell holdouts for its nonlinear attribution model and paired selective-mask contrasts. The controlled bridge uses identity-matched donors. All negative, failed, and near-chance cells remain in machine-readable artifacts even when excluded from positive mechanistic interpretation.

D Signal, fluctuation, and repair derivation

For realization i, write the correct-target margin as $m_{li} = S_l + \eta_{li}$ with $S_l = \mathbb{E}_i[m_{li}]$ and $N_l^2 = \mathrm{Var}_i(m_{li})$. A scale-free diagnostic is $\Gamma_l = S_l/\sqrt{N_l^2 + \epsilon}$. With layer update $\Delta m_{li} = \mu_l + \varepsilon_{li}$,

$$m_{Li} = m_{0i} + \sum_l \mu_l + \sum_l \varepsilon_{li},$$

and therefore

$$\mathrm{Var}\left(\sum_l \varepsilon_l\right) = \sum_l \sigma_l^2 + 2\sum_{l<j} \mathrm{Cov}(\varepsilon_l, \varepsilon_j).$$

Negative off-diagonal covariance is an operational signature of later repair: later updates cancel part of earlier realization-specific deviation. The reported summed off-diagonal term is $2\sum_{l<j} \mathrm{Cov}(\varepsilon_l, \varepsilon_j)$ and already includes the factor of two. It does not imply that every layer contracts variance, nor that total hidden-state variance must shrink.

For a paired displacement, local linearization gives

$$\delta r_{l+1} = J_l \delta r_l + O(\|\delta r_l\|^2).$$

The frozen cumulative approximation is $\widehat{\delta r_b} = J_{b-1} \cdots J_a \delta r_a$. It is evaluated by directional cosine, relative norm error, and decision-margin projection. Piecewise re-linearization recomputes the Jacobian along the perturbed path. Controlled results show strong one-step directional agreement but degrading cumulative fidelity, so the Jacobian is a local diagnostic rather than a globally linear explanation.

E Full head analyses

Zero ablation, mean replacement, equivalent-family replacement, mismatched replacement, and pair interventions separate generic component magnitude from family-specific causal utility. Equivalent replacement is least damaging in the original controlled scan, while mismatch produces greater discrimination damage. Pair utilities are non-additive. Attention-motif recurrence does not predict causal head utility in either the original or consolidated scan. Nested target-preserving comparisons show partial top-head overlap and continued recruitment, not a stable semantic head inventory.

F Nested-pattern details

The causal bridge crosses irrelevant, supportive, refining, override, and multilevel extensions with matched embedded-short, unrelated-same-target, and nonequivalent donors. Only target-preserving strata clear competence. Embedded-short replacement is less damaging than nonequivalent replacement, supporting partial reuse of a learned increment. Target-changing hierarchy and override variants remain failure diagnostics. The later length study likewise restricts internal interpretation to competent cells.

G Predictive-order phase grid

The complete tracked tables are `phase_grid_results.csv`, `phase_min_depth.csv`, `phase_min_width.csv`, `phase_internal_decision_depth.csv`, and `phase_model_fits.csv`. They report every trained depth/width/budget/order cell and preserve failed high-order attempts. Multiple surface lengths, three dependency spans, and four nuisance counts are averaged only after their matched cell results are retained.

H Matched stress frontier and budget audit

The confirmatory stress design retains 29 axis-specific contrast rows over 25 unique physical dataset cells. Shared baseline examples appear in every relevant one-factor contrast but are deduplicated during training. L2/W64/H4 has 85,504 parameters and L8/W32/H4 has 77,632, a 9.2% gap. Three model seeds are evaluated at each exact-prefix training budget. The 2x and 4x runs use the same initialization and respectively extend the first 400 and 800 training batches, so budget rescue is not confounded by a fresh draw.

Architecture	Seed	1x	2x	4x
L2/W64/H4	11	.188	.906	1.000
	23	.250	1.000	1.000
	37	1.000	1.000	1.000
L8/W32/H4	11	1.000	1.000	1.000
	23	.375	1.000	1.000
	37	.312	.750	1.000

Table 2: Held-out order-two baseline accuracy under exact-prefix budget scaling. Factor slopes are interpreted only at 4x, where all six cells cross the 0.80 gate.

At 4x both architectures have $p^*_{\max} = 2$, $n_{\max} = 24$, $k_{\max} = 32$, and $s_{\max} = 32$ over the tested grid. The latter three are right-censored lower bounds because every maximum cell remains competent. The ten three-seed failures among 58 architecture/contrast cells are precisely the five $p^* \geq 3$ cells for each architecture. Machine-readable seed, aggregate, trajectory, loss, and frontier tables are under `results/stress_frontier/learned_v1`, `learned_v2_matched`, and `learned_v4_matched`.

An earlier diagnostic directory named `learned_v2` changed initialization and data-order seeds with budget. It is explicitly marked `INVALID_BUDGET_COMPARISON.md`, is excluded from every table and figure, and supports no claim.

I Recurrent comparison details

R1 fixes $p^* = 2$ and varies raw length; R2 fixes $p^* = 2$ and varies the distance between the two indispensable values; R3 varies predictive order at controlled span; R4 varies same-vocabulary nuisance. Recurrent target margin and hidden norm are sampled through time. Matched nuisance pairs additionally record $\|h_t^{(k)} - h_t^{(0)}\|_2$. Transformer margins are decoded at depth boundaries. These plots diagnose different axes and do not equate a recurrent transition with a Transformer block.

J Pretrained Pythia and Qwen diagnostics

Behavioral filtering retains 10 of 12 equal-length pairs across pinned Pythia-70M and Qwen3-0.6B checkpoints. Mean cumulative directional cosine is 0.768, 0.733, and 0.702 over one, two, and four blocks, but relative-norm error exceeds one in several cells. The broader replacement matrix finds equivalent/nonequivalent contrasts, while an on-manifold nearest-donor control sharply weakens a label-specific interpretation. These are external-validity diagnostics, not evidence that the controlled equivalence classes occur naturally at scale.

K Legacy pilot and continual-learning boundary

The preserved pilot contains the original atlas, component interventions, motif controls, entropy trajectories, equivalence patches, and training checkpoints. Familiar low-entropy mappings are FF-heavier under zero ablation, whereas contextual override/repair is SA-heavier. Controlled frequency/mechanism regression is weak. These results motivate distributed, state-dependent computation but are secondary to the central refinement theory.

Paper 0.5 neither implements nor validates continual learning. Stable predictive equivalence is at most a diagnostic precursor. Persistent prototypes or adapters remain blocked pending natural held-out causal utility, matched-budget benefit, contamination controls, and rollback.

L Reproduction commands

```
PYTHONPATH=src python -m cl.experiments.paper05_predictive_order_phase
PYTHONPATH=src python -m cl.analysis.paper05_predictive_order_phase
PYTHONPATH=src python -m cl.experiments.paper05_stress_run \
  --device mps --tranche rescue_t4 --resume
PYTHONPATH=src python -m cl.experiments.paper05_stress_analyze \
  --input results/stress_frontier/learned_v4_matched
PYTHONPATH=src python -m cl.experiments.paper05_stress_figures
```

```
PYTHONPATH=src python -m cl.experiments.paper05_rnn_comparison
PYTHONPATH=src python -m cl.analysis.paper05_rnn_comparison
PYTHONPATH=src python -m cl.experiments.paper05_layer_composition
PYTHONPATH=src python -m cl.experiments.paper05_group25_sa_variance
PYTHONPATH=src python -m cl.experiments.paper05_nested_patterns
```

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